PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL

SONGEA DISTRICT COUNCIL

FORM FOUR MOCK EXAMINATION

032/2A **CHEMISTRY2A (ACTUAL PRACTICAL)**

(For Both School and Private Candidates)

Time: 3 Hours **March, 2025**

Instructions

1. Write your examination number on every page of your answer sheet provided.
2. Answer ALL Questions
3. Scientific calculators and graph may be used
4. The following constant may be used

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H = 1, Cl = 35.5, C = 12, O = 16, S = 32

1. You are provided with the following solutions;

A: Containing 0.1 mole hydrochloric acid per dm^3 of solution;

B: Contain 2.65g M_2CO_3 per 0.5dm^3 of solution

Methyl orange indicator.

Questions;

a. Titrate G (in burette) against B (in conical flask) using two drops of your indicator obtain three titre values. Record your data in a tabular form.

(i) Cm3····· of B required …….Cm^3 of G for complete reaction.

ii. Write a balanced chemical equation for the reaction between B and G and the

corresponding ionic equation with state symbols.

iii. Showing your procedures clearly; calculate the molar mass of M_2CO_3 and hence identify element M

2. You are provided with the following:

X: A solution containing 79g of $Na_2S_2O_3.5H_2O$

Y: A solution contain 1.0 mol/dm^{-1} of dilute HCL

T: Distilled water

Stop watch/clock

Procedure

(i) Place a 100cm^3 beaker on top of letter X on a plain paper provided

(ii) Measure 8.0cm^3 of U and 2cm^3 of T and put them in the beaker in (i)

(iii) Measure 10cm^3 of V and put it into a beaker containing U and T; immediately start the stop watch and observe the changes from above

(iv) Record the time taken for the disappearance of letter X

(v) Repeat steps (i) to (iv) using the data shown in table 1.

Table 1

Number of experiment	Volume of U(cm^3)	Volume of V(cm^3)	Volume of T(cm^3)	Time (t) in second	1/t(sec^{-1})
1	8	10	2		
2	6	10	4		
3	4	10	6		
4	2	10	8		
5	1	10	9		

Questions:

a) What is the aim of the whole experiment

b) Complete table 1

c) Give reasons (s), identify the experiment in which the reaction was

i. Fast

ii. Slow

d) With state symbols write balanced chemical equation for the reaction between U and V

e) List four factors which can affect rate of chemical reaction

f) Write the electronic configuration of the production which causes the solution to cloud letter X

g) What can you conclude from the data obtain